# Boyce–Codd Normal Form (BCNF)

## 1 Introduction

**Definition: BCNF**

A relation schema $R$ is in **Boyce–Codd Normal Form (BCNF)** with respect to a set of functional dependencies $F$ if, for every functional dependency $\alpha \to \beta$ in $F^+$, where $\alpha, \beta \subseteq R$, at least one of the following holds:

1. $\alpha \to \beta$ is **trivial**, i.e., $\beta \subseteq \alpha$.
2. $\alpha$ is a **superkey** of $R$.

**Key Idea**

BCNF eliminates *all redundancy* that can be discovered via functional dependencies. However, other forms of redundancy (e.g., multivalued dependencies) may still remain.

## 2 Example: Schema Not in BCNF

**Example**

Consider:

$$\text{inst\_dept(ID, name, salary, dept\_name, building, budget)}$$

Given:

$$\text{dept\_name} \to \text{building, budget}$$

Here, dept_name is *not* a superkey, so BCNF is violated.

**BCNF Decomposition**

We decompose inst_dept into:

$$\text{department(dept\_name, building, budget)}$$

$$\text{instructor(ID, name, salary, dept\_name)}$$

Both relations are in BCNF.

# 3 BCNF Decomposition Process

**BCNF Decomposition Algorithm**

1. Start with schema $R$ and $F$.
2. While there exists a violation:
   (a) Find a non-trivial dependency $\alpha \to \beta$ in $F^+$ such that:
   - $\alpha$ is not a superkey.
   - $\beta \not\subseteq \alpha$.

   (b) Replace $R$ with:
$$R_1 = \alpha \cup \beta$$
$$R_2 = R - (\beta - \alpha)$$
3. Repeat until all schemas are in BCNF.

# 4 Testing for BCNF

**Testing Steps**

1. For each non-trivial FD $\alpha \to \beta$:
   - Compute $\alpha^+$.
   - If $\alpha^+$ contains all attributes of $R$, $\alpha$ is a superkey (no violation).
   - Otherwise, BCNF is violated.
2. Only checking $F$ may be enough for original schema, but *not* for decomposed relations — you may need $F^+$.

# 5 Worked Examples

## Example 1: Simple BCNF Decomposition

**Simple Example**

Relation:
$$R = (A, B, C), \quad F = \{A \to B,\ B \to C\}$$
Key = $\{A\}$.
Violation: $B \to C$ (B is not a superkey).
**Decomposition:**
$$R_1 = (B, C), \quad R_2 = (A, B)$$
Both $R_1$ and $R_2$ are in BCNF.

## Example 2: Classroom Example

**Classroom Example**

Relation:

class(course_id, title, dept_name, credits, sec_id, semester, year,

building, room_number, capacity, time_slot_id)

FDs:

course_id $\rightarrow$ title, dept_name, credits

building, room_number $\rightarrow$ capacity

course_id, sec_id, semester, year $\rightarrow$ building, room_number, time_slot_id

Candidate key: {course_id, sec_id, semester, year}.

**Decomposition:**

course(course_id, title, dept_name, credits)

classroom(building, room_number, capacity)

section(course_id, sec_id, semester, year, building, room_number, time_slot_id)

All resulting relations are in BCNF.

## Example 3: Department-Project Assignment

**Department-Project Assignment**

Relation:

$$\text{assign}(\text{emp\_id}, \text{emp\_name}, \text{dept\_id}, \text{dept\_name}, \text{proj\_id}, \text{proj\_name})$$

FDs:

$$\text{emp\_id} \to \text{emp\_name}, \text{dept\_id}$$

$$\text{dept\_id} \to \text{dept\_name}$$

$$\text{proj\_id} \to \text{proj\_name}$$

Candidate key: {emp_id, proj_id}.
Violations:

- dept_id → dept_name (dept_id is not a superkey).
- proj_id → proj_name (proj_id is not a superkey).

**Decomposition:**

$$\text{employee}(\text{emp\_id}, \text{emp\_name}, \text{dept\_id})$$

$$\text{department}(\text{dept\_id}, \text{dept\_name})$$

$$\text{project}(\text{proj\_id}, \text{proj\_name})$$

$$\text{assignment}(\text{emp\_id}, \text{proj\_id})$$

All resulting relations are in BCNF.

## Example 4: Product-Supplier

**Product-Supplier Example**

Relation:

$$\text{product\_supply(prod\_id, prod\_name, supp\_id, supp\_name, supp\_city)}$$

FDs:

$$\text{prod\_id} \rightarrow \text{prod\_name}$$

$$\text{supp\_id} \rightarrow \text{supp\_name, supp\_city}$$

Candidate key: {prod_id, supp_id}.
Violations:

- prod_id → prod_name (prod_id not superkey).
- supp_id → supp_name, supp_city (supp_id not superkey).

**Decomposition:**

$$\text{product(prod\_id, prod\_name)}$$

$$\text{supplier(supp\_id, supp\_name, supp\_city)}$$

$$\text{product\_supplier(prod\_id, supp\_id)}$$

All resulting relations are in BCNF.

## Example 5: BCNF Decomposition with $\alpha \rightarrow \beta$

**BCNF Decomposition Example**

Given relation schema:

$$r(\alpha, \beta, \gamma)$$

and functional dependency:

$$\alpha \rightarrow \beta,$$

we decompose $r$ into:

$$r_1(\alpha, \beta) \quad \text{and} \quad r_2(\alpha, \gamma).$$

**a. Expected Constraints:**

- $\alpha$ is the **primary key** in both $r_1$ and $r_2$.
- $\alpha$ in $r_2$ acts as a **foreign key** referencing $\alpha$ in $r_1$.

**b. Example of Inconsistency Without Foreign-Key Enforcement:**

- Suppose $r_1$ has tuples with $\alpha = a_1, a_2$.
- Without foreign key enforcement, $r_2$ could have a tuple with $\alpha = a_3$ that does not exist in $r_1$.
- This leads to an inconsistency because $r_2$ references a nonexistent $\alpha$.

**c. 3NF Decomposition Constraints:**

- Each decomposed relation has a **primary key** consisting of the determinant $X$ from an FD $X \to A$.
- **Foreign keys** link relations to preserve referential integrity.
- 3NF decomposition preserves dependencies and lossless join but may retain some redundancy.

## Example 6: BCNF Decomposition with $\alpha \to \beta$

**BCNF Decomposition Example**

Relation schema:

$$R = (A, B, C, D, E)$$

Functional dependencies:

$$F = \{A \to BC, \quad CD \to E, \quad B \to D\}$$

Decomposition:

$$R_1 = (A, B, C), \quad R_2 = (A, D, E)$$

# Part 1: Show the decomposition is lossless

### Step 1: Find common attributes

$$R_1 \cap R_2 = \{A\}$$

### Step 2: Lossless join condition

The decomposition $R_1, R_2$ is lossless if:

$$(R_1 \cap R_2) \to R_1 \quad \text{or} \quad (R_1 \cap R_2) \to R_2$$

i.e., the common attributes functionally determine all attributes of at least one component.

### Step 3: Compute closure $A^+$ under $F$

$$\begin{aligned} A &\xrightarrow{A \to BC} \{A, B, C\} \\ &\xrightarrow{B \to D} \{A, B, C, D\} \\ &\xrightarrow{CD \to E} \text{Since } C, D \in \{A, B, C, D\}, \Rightarrow \{A, B, C, D, E\} \end{aligned}$$

Thus,

$$A^+ = \{A, B, C, D, E\} = R$$

### Conclusion

Since $A$ (the intersection) functionally determines all attributes in $R$, it certainly determines all attributes in both $R_1$ and $R_2$. Hence, the decomposition is **lossless**.

$$\boxed{\text{Decomposition } R = R_1 \cup R_2 \text{ is lossless}}$$

## Part 2: Find a lossless BCNF decomposition of $R$

### Step 1: Check BCNF violations in $R$

Check each FD $\alpha \to \beta$ for $\alpha$ being a superkey in $R$:

- $A \to BC$: $A^+ = R$ (shown above), so $A$ is a key. **No violation.**
- $CD \to E$: Compute $(CD)^+$:

  $$CD \xrightarrow{CD \to E} \{C, D, E\}$$

  Does not contain all attributes of $R$, so $CD$ is **not a key**. **Violation!**
- $B \to D$: Compute $B^+$:

  $$B \xrightarrow{B \to D} \{B, D\}$$

  Not all attributes $\to$ **Violation!**
- $E \to A$: Compute $E^+$:

  $$E \xrightarrow{E \to A} \{E, A\} \xrightarrow{A \to BC} \{A, B, C, E\} \xrightarrow{B \to D} \{A, B, C, D, E\} = R$$

  $E$ is a key $\to$ **No violation.**

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### Step 2: Decompose on violation $CD \to E$

Decompose $R$ into:

$$R_3 = (C, D, E), \quad R_4 = (A, B, C, D)$$

- In $R_3$, $CD \to E$, and $CD$ is a candidate key for $R_3$. **BCNF holds.**

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### Step 3: Check $R_4$ for BCNF

Functional dependencies projected on $R_4$:

$$F_{R_4} = \{A \to BC, \quad B \to D\}$$

Check keys for $R_4$:
- Try $AB$:

$$(AB)^+ = \{A, B, C, D\} = R_4$$

So candidate key is $AB$.

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Check violations:
- $A \to BC$: $A$ alone is not a superkey → **violation**. - $B \to D$: $B$ alone is not a superkey → **violation**.

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### Step 4: Decompose $R_4$ on $A \to BC$

$$R_5 = (A, B, C), \quad R_6 = (A, B, D)$$

- $R_5$: $A \to BC$, $A$ is a key → **BCNF holds**.

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### Step 5: Check $R_6$ for BCNF

FD on $R_6$:

$$B \to D$$

Candidate keys in $R_6$:
- $AB$:

$$(AB)^+ = \{A, B, D\} = R_6$$

Violation because $B \to D$ and $B$ is not a key.

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### Step 6: Decompose $R_6$ on $B \to D$

$$R_7 = (B, D), \quad R_8 = (A, B)$$

- $R_7$: $B \to D$, $B$ is key → **BCNF holds**. - $R_8$: No non-trivial FDs → **BCNF holds**.

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## Final BCNF Decomposition

$$\boxed{\begin{cases} R_3 = (C, D, E), & CD \to E \\ R_5 = (A, B, C), & A \to BC \\ R_7 = (B, D), & B \to D \\ R_8 = (A, B) \end{cases}}$$

Each relation is in BCNF, and the decomposition is lossless.

# 6 Dependency Preservation in BCNF

**Important Note**

It is *not always possible* to have a BCNF decomposition that is dependency preserving.
Example:

$$R = (J, L, K), \quad F = \{JK \to L,\ L \to K\}$$

Candidate keys: $JK, JL$.
No BCNF decomposition can preserve all dependencies without joins.

# 7 More Examples: Design Goal

This section explores various functional dependency (FD) scenarios for a relation $R = (A, B, C)$, examining the **candidate key**, **decomposition**, and whether the decomposition satisfies:

- Lossless-Join property
- BCNF
- Dependency Preservation

**Legend**

- ✓ : Property satisfied
- NA : Not applicable (no decomposition needed)
- "No" : Property not satisfied

| Sl | FDs | Key | Decomposition | Lossless? | BCNF? | Dependency Preserving? |
|---|---|---|---|---|---|---|
| 1 | $A \to B$, $A \to C$ | $A$ | No decomposition needed | NA | Already | NA |
| 2 | $A \to B$, $B \to C$ | $A$ | $R_1 = (B, C)$, $R_2 = (A, B)$ | ✓ | ✓ | ✓ |
| 3 | $A \to B$, $C \to A$ | $C$ | $R_1 = (A, B)$, $R_2 = (A, C)$ | ✓ | ✓ | ✓ |
| 4 | $A \to B$, $C \to B$ | $AC$ | $R_1 = (A, B)$, $R_2 = (A, C)$ | ✓ | ✓ | No (misses $C \to B$) |
| | | | $R_1 = (B, C)$, $R_2 = (A, C)$ | ✓ | ✓ | No (misses $A \to B$) |
| 5 | $A \to C$, $B \to A$ | $B$ | $R_1 = (A, C)$, $R_2 = (A, B)$ | ✓ | ✓ | ✓ |
| 6 | $A \to C$, $C \to B$ | $A$ | $R_1 = (B, C)$, $R_2 = (A, C)$ | ✓ | ✓ | ✓ |
| 7 | $A \to C$, $B \to C$ | $AB$ | $R_1 = (A, C)$, $R_2 = (A, B)$ | ✓ | ✓ | No |
| | | | $R_1 = (B, C)$, $R_2 = (A, B)$ | ✓ | ✓ | No |
| 8 | $B \to A$, $B \to C$ | $B$ | No decomposition needed | NA | Already | NA |
| 9 | $B \to A$, $C \to A$ | $BC$ | $R_1 = (A, B)$, $R_2 = (B, C)$ | ✓ | ✓ | No |
| | | | $R_1 = (A, C)$, $R_2 = (B, C)$ | ✓ | ✓ | No |
| 10 | $B \to A$, $C \to B$ | $C$ | $R_1 = (A, B)$, $R_2 = (B, C)$ | ✓ | ✓ | ✓ |
| 11 | $B \to C$, $C \to A$ | $B$ | $R_1 = (A, C)$, $R_2 = (B, C)$ | ✓ | ✓ | ✓ |
| 12 | $C \to A$, $C \to B$ | $C$ | No decomposition needed | NA | Already | NA |

## Key Observations

- If one attribute functionally determines all others, no decomposition is needed (Cases 1, 8, 12).
- Some decompositions can achieve BCNF and lossless-join but fail dependency preservation (e.g., Cases 4, 7, 9).

- When multiple decompositions are possible, check both for dependency preservation (Cases 4, 7, 9).
- BCNF does not guarantee dependency preservation — a common exam trick.

**Exam Tip**

In questions where BCNF decomposition is required, always check:

1. **Lossless-join** — ensure $(R_1 \cap R_2) \to R_1$ or $(R_1 \cap R_2) \to R_2$ holds.
2. **Dependency preservation** — verify that all original FDs can be enforced without join.